

STARTUP SCREENING REPORT

TerraSpark

Independent Public-Source Assessment

Founder Screen

www.terraspark.energy

RESEARCH CUT-OFF

30 March 2026

REPORT DATE

28 September 2026

REPORT REFERENCE

f2a8fd552eed

Key facts & context

BASE COUNTRY **Luxembourg** FOUNDED **2025** STAGE **Pre-Seed** SECTOR **Deeptech · space solar** CUSTOMER MODEL **B2B · enterprise**

WHAT TERRASPARK DOES

TerraSpark is developing space-based solar power systems. Its offering includes space-based solar power systems and wireless power transmission systems. It targets data center operators, critical infrastructure operators and defence installations, among other segments, with Luxembourg and Europe as named markets. The revenue model is project-based, sold to enterprise customers.

CURRENT POSITION

TerraSpark is a Pre-Seed-stage company, founded in 2025, based in Luxembourg. In March 2026 it stated funding would support initial pilot applications and live tests of its space-based solar technology. No customer commitment has been disclosed yet; the commercial case rests on the product and the need it addresses. It has completed one financing in 2026. The company is led by its three-person founding team (CEO).

COMPANY PROFILE

Founded	2025	Headquarters	Luxembourg
Stage	Pre-Seed	Sector	Deeptech
Sub-sector	Space solar	Customer model	B2B, enterprise customers, project bid
Revenue model	Project-based, non-recurring	ICP	Chief technology, infrastructure, energy, or sustainability executives at data centers, utilities, critical infrastructure operators, defence organizations, mining companies, and desalination operators
Target customers	Data center operators, critical infrastructure operators, defence installations, mining companies and water desalination plants	Core technology	Space systems and wireless power transmission
Website	www.terraspark.energy		

PRODUCTS & SERVICES

- 1 Space-based solar power systems
- 2 Wireless power transmission systems

COMPANY TIMELINE

Date	Event
2025	Company founded.
March 2026	March 2026: TerraSpark stated funding would support initial pilot applications and live tests of its space-based solar technology.
March 2026	March 2026: TerraSpark was reported to be targeting an orbital technology demonstrator in 2027 and first space-to-Earth power transmission in 2028.
March 2026	Pre-Seed round, EUR 5m.

TARGET MARKETS

- 1 Luxembourg
- 2 Europe

LEGAL & REGULATORY

Material legal or regulatory proceedings
None identified.

INFORMATION TO PREPARE

- 1. The registered legal entity and its jurisdiction.

Founder & team

BASE COUNTRY Luxembourg FOUNDED 2025 STAGE Pre-Seed SECTOR Deeptech · space solar CUSTOMER MODEL B2B · enterprise

Founding team	3 - two described as co-founders (Sanjay Vijendran and Matthias Laug); one described as a founder (Jasper Deprez)
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FOUNDER DETAILS

Jasper Deprez CEO · Founder

Education	Master of Business Administration, Commerce — Esei International Business School Barcelona (2015)
Past startup experience	Founder of Tradler App (2017–present). No exit or acquisition was publicly identified
Other current roles	Founder, Tradler App (2017–present)

FOUNDER–MARKET FIT

Deprez has co-founded or led other ventures (Founder of Tradler App. No exit or acquisition was publicly identified), experience relevant to building an early-stage company. Formal training in Administration complements this.

Sanjay Vijendran Co-founder

Education	Doctor of Philosophy in Physics, University of Cambridge (2001); Doctorates, University of Cambridge (2001); Bachelor of Science in Physics, Imperial College London (1996); The Leys School
Experience	Watch This Space, European Space Agency (2024–2025); Lead for ESA's Solaris Initiative on Space-Based Solar Power, European Space Agency (2021–2024); Future Mars Studies Coordinator, European Space Agency (2020–2022); ESA Mars Sample Return Lead, European Space Agency (2016–2020); Applied Physicist, European Space Agency (2009–2015)
Other current roles	Director and Principal Consultant, Space Energy Insights (2025–present)

FOUNDER–MARKET FIT

Sanjay Vijendran's time at European Space Agency and Space Energy Insights gave experience in the space and aerospace and energy sectors, directly relevant to a company building developing space-based solar power systems. Vijendran's prior roles span research. Formal training in Physics complements this.

Matthias Laug Co-founder

Experience	CPTO, Enter (2023–2025); Chief Technology Officer and Co-Founder, Tier Mobility (2018–2023); Lead Software Engineer, Thoughtworks (2016–2018); Chief Technology Officer, takeaway.com (2014–2016); Chief Technology Officer, Yd Yourdelivery (2009–2014)
Past startup experience	Chief Technology Officer and Co-Founder of Tier Mobility (2018–2023). No exit or acquisition was publicly identified

FOUNDER–MARKET FIT

Matthias Laug's prior roles span engineering. Laug has co-founded or led other ventures (Chief Technology Officer and Co-Founder of Tier Mobility. No exit or acquisition was publicly identified), experience relevant to building an early-stage company.

TEAM–MARKET FIT

The founding team combines sector experience from European Space Agency and Space Energy Insights with research and engineering backgrounds. For a Pre-Seed-stage software company, the roles and prior experience visible today cover product engineering. Prior experience in sales and go-to-market, customer success and finance is not visible from the founders' backgrounds and should be clarified.

MANAGEMENT COVERAGE

The three-person founding team holds the senior roles, covering executive leadership; a founder-led structure is usual at this stage. As the company scales, execution depends increasingly on repeatable sales, customer success and engineering management. The depth of management below the founders in these functions should be clarified with management.

INFORMATION TO PREPARE

1. Whether Jasper Deprez and Sanjay Vijendran work full-time for the company, given the other current roles listed.
2. Current headcount by function and the management responsibilities below the founders.
3. Board and advisory composition.
4. Key hires planned for the next 12-18 months and the functions they cover.

Product & technology

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WHAT TERRASPARK IS DEVELOPING

It is developing developing space-based solar power systems, combining space-based solar power systems and wireless power transmission systems into one offering.

PRODUCT PORTFOLIO

#	Product / component	What it does	Primary use
1	Space-based solar power systems	-	-
2	Wireless power transmission systems	-	-

POSITION IN THE VALUE CHAIN

TerraSpark operates in upstream R&D and system integration, developing orbital generation and wireless transmission systems before delivering power to enterprise customers.

PRODUCT & TECHNOLOGY DIFFERENTIATION

What may be harder for alternatives to replicate is proprietary technology / know-how and system integration, which depend on execution rather than on protected technology alone.

CURRENT PRODUCT MATURITY

In March 2026 it stated funding would support initial pilot applications and live tests of its space-based solar technology. March 2026: TerraSpark was reported to be targeting an orbital technology demonstrator in 2027 and first space-to-Earth power transmission in 2028. The next meaningful step is use by paying customers.

CORE TECHNOLOGY

The system combines four elements: orbital solar arrays capture sunlight; radio-frequency transmitters wirelessly deliver power; remote receivers or rectennas convert radiowaves into electricity; satellites, ground infrastructure, and control systems integrate the end-to-end system. According to the company: develops modular orbital solar generation and precisely directed RF wireless power transmission.

WHY NOW?

Cheaper launches and reusable spacecraft enable orbital demonstrations; example: planned 2027 demonstrator. TerraSpark's offering is built on this progress.

INTELLECTUAL PROPERTY

No patent position is established for TerraSpark; whether its software, data and algorithms are protected by patents, held as trade secrets or owned jointly with research partners requires confirmation.

INFORMATION TO PREPARE

1. Ownership of intellectual property, including results developed with research partners.
2. The next technical milestone and its expected date.

Validation & market signals

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VALIDATION POSITION

No customer commitment has been disclosed yet; TerraSpark's external validation so far comes from research and programme partnerships. The next transition is a first customer commitment - a pilot, letter of intent or order.

STATED PIPELINE

The company reports preparation for initial pilot applications and live tests; counterparties are not named. These are plans the company describes, not commitments.

COMMERCIAL TRACTION

No commercial metrics have been disclosed; for a Pre-Seed-stage software company the relevant measures are customers, ARR or MRR, retention and expansion.

EXTERNAL VALIDATION & STRATEGIC ENGAGEMENT

Organisation	Nature of engagement	Relevance
Dcubed	Dcubed announced TerraSpark among five power-beaming partners for the Araqys D3 mission.	External engagement; the commercial role is to be confirmed.

NEXT VALIDATION MILESTONE

A first paying customer or a paid pilot.

INFORMATION TO PREPARE

1. Customer engagements to date: pilots, letters of intent or orders, with counterparty type and status.

Market opportunity

BASE COUNTRY Luxembourg FOUNDED 2025 STAGE Pre-Seed SECTOR Deeptech · space solar CUSTOMER MODEL B2B · enterprise

MARKET DEFINITION

TerraSpark operates in the emerging space-based solar power and wireless-energy market, combining orbital solar generation, radio-frequency transmission, and enterprise energy infrastructure for continuous power delivery. The direct market is the operators who have this need - data center operators, critical infrastructure operators, defence installations and mining companies.

TARGET CUSTOMER & BUYER

Target customers	Data center operators, critical infrastructure operators, defence installations, mining companies and water desalination plants
ICP	Chief technology, infrastructure, energy, or sustainability executives at data centers, utilities, critical infrastructure operators, defence organizations, mining companies, and desalination operators
Economic buyer & budget	A data-centre, infrastructure, defence, mining, or desalination executive signs from energy-infrastructure capital expenditure when grid reliability, access, or resilience is inadequate
End users	Facility operators, energy-management teams, infrastructure teams, and personnel responsible for power continuity at remote or critical sites
Buying trigger	Grid unreliability, lack of transmission infrastructure, high energy costs in remote locations, disaster-response requirements, or demand for weather-independent power resilience
Target markets	Luxembourg and Europe

INITIAL MARKET WEDGE

Luxembourg, the home market, is the first named market; Europe is also named. The company has not stated which segment and geography it will prioritise, or why.

DEMAND DRIVERS

- Demand: remote sites need reliable power where grids or fuel logistics are costly; example: mines and island grids.
- Resilience: critical operators seek weather-independent electricity; example: disaster-response and defence applications.

ADOPTION BARRIERS

- High launch and orbital-system costs, unproven large-scale economics, spectrum and safety regulation, receiver and grid infrastructure requirements, and competition from terrestrial renewables, batteries, nuclear, and diesel generation.

MARKET SIZE & ECONOMICS

The company sells Power-as-a-Service based on delivered wireless electricity; current commercial model is not yet fully established, with terrestrial pilots preceding orbital service deployment. Bottom-up, the direct opportunity is addressable customers × projects per customer per year × average project value; these inputs have not been disclosed, so no market size is stated.

MARKET ACCESSIBILITY

Buyers are identifiable: operators of data center operators, critical infrastructure operators, defence installations and mining companies. Budget ownership: a data-centre, infrastructure, defence, mining, or desalination executive signs from energy-infrastructure capital expenditure when grid reliability, access, or resilience is inadequate. Access is constrained by high launch and orbital-system costs, unproven large-scale economics, spectrum and safety regulation, receiver and grid infrastructure requirements, and competition from terrestrial renewables, batteries, nuclear, and diesel generation, which lengthen the sales cycle.

MARKET OPPORTUNITY - ANALYST VIEW

The opportunity is supported by clear demand drivers - remote sites need reliable power where grids or fuel logistics are costly, and critical operators seek weather-independent electricity. It is constrained by high launch and orbital-system costs, unproven large-scale economics, spectrum and safety regulation, receiver and grid infrastructure requirements, and competition from terrestrial renewables, batteries, nuclear, and diesel generation. Before scale can be judged, the number of addressable customers, price and unit economics and evidence that customers commit need to be established.

INFORMATION TO PREPARE

1. Identify the initial customer segment, use case and geography the company is prioritising, and why that one first.
2. Provide the sizing inputs - (1) addressable customers; (2) projects per customer per year (for offerings such as Space-based solar power systems, Wireless power transmission systems); (3) average project value - each with its source and the date it was measured. Any existing market sizing, with its figures, method and sources.
3. Describe the buying process: decision-makers, budget owner, and typical cycle length from first contact to contract.

Competitive landscape

BASE COUNTRY Luxembourg FOUNDED 2025 STAGE Pre-Seed SECTOR Deeptech · space solar CUSTOMER MODEL B2B · enterprise

COMPETITIVE STRUCTURE

Customers who need to provide reliable continuous electricity to hard-to-reach operations without dependable terrestrial grid access choose between continuing with their current way of working (terrestrial grid connection and diesel generation), adjacent automated systems (Aetherflux), like-for-like systems (Space Solar), extensions of incumbent suppliers' offerings (Rolls-Royce) and in-house builds or systems-integrator projects (Thales Alenia Space). Space Solar offers a directly comparable solution.

KEY COMPETITORS & ALTERNATIVES

Alternative	Category	Offering	Why it is in the comparison
Terrestrial grid connection and diesel generation	The customer's current way of working	Grid connection and diesel backup generation.	It would take the remote-site power decision by offering an established terrestrial supply solution instead of TerraSpark's proposed orbital supply.
Aetherflux	Adjacent automated systems	Space-based solar power and wireless power transmission.	Aetherflux could take strategic customers and capital evaluating orbital power generation for resilient terrestrial electricity.
Space Solar	Like-for-like systems	Space-based solar power system development.	Space Solar could capture European utility, infrastructure, or government attention for the same future orbital-electricity procurement.
Rolls-Royce	Extensions of incumbent suppliers' offerings	Distributed power generation systems.	Rolls-Royce could retain an industrial or defence account by extending conventional power-generation infrastructure rather than adopting TerraSpark's orbital system.
Thales Alenia Space	In-house builds or systems-integrator projects	Space systems and infrastructure integration.	A government or utility-led programme could commission Thales Alenia Space to assemble a bespoke orbital power system instead of selecting TerraSpark.

HOW CUSTOMERS MAY CHOOSE

Criterion	TerraSpark - established position
Performance / throughput Delivered power capacity and transmission efficiency determine suitability for demanding operations	Plans orbital solar collection and RF transmission to ground receiving stations.
Reliability / uptime Continuous electricity is central where grid access or resilience is inadequate	Claims clean, reliable, 24/7 power independent of weather.
Infrastructure requirement Ground receiving equipment, generation assets, and site works affect deployment feasibility	Requires orbital solar arrays and ground rectenna receiving stations connected to the grid.
Development stage Customers must assess whether the technology is commercially available or remains demonstrational	Phased development begins with terrestrial wireless transmission before planned orbital commercialization.

Comparable evidence on Terrestrial grid connection and diesel generation, Aetherflux, Space Solar, Rolls-Royce and Thales Alenia Space has not been established for these criteria. Upfront cost and operating economics are not established for any party.

WHAT COULD BECOME HARDER TO REPLICATE

Criterion	TerraSpark - established position
Proprietary technology / know-how Specialized space-generation and RF transmission know-how could be difficult to reproduce	Develops modular orbital solar generation and precisely directed RF wireless power transmission.
System integration Integrating orbital generation, wireless transmission, receivers, and grid delivery could protect position	System combines orbital solar arrays, RF transmission, rectennas, and grid delivery.

Comparable evidence on Terrestrial grid connection and diesel generation, Aetherflux, Space Solar, Rolls-Royce and Thales Alenia Space has not been established for these factors. Intellectual property, ecosystem / partnerships, scale and capital and regulatory approvals / certification are not established for any party.

COMPETITIVE POSITION

TerraSpark's comparative position rests on performance / throughput, reliability / uptime, infrastructure requirement and development stage. The trade-off is site infrastructure (requires orbital solar arrays and ground rectenna receiving stations connected to the grid), which mobile and manual alternatives do not need. Relative upfront cost and operating economics are not established. The company's cost and performance figures are its own until supported by operating data.

DEFENSIBILITY - EXISTING SOURCES

- Proprietary technology / know-how: develops modular orbital solar generation and precisely directed RF wireless power transmission.
- System integration: system combines orbital solar arrays, RF transmission, rectennas, and grid delivery.

DEFENSIBILITY - EMERGING / POTENTIAL SOURCES

- Customer base and usage data: these build only with paying customers.
- Protected technology: no patent position is established, so defensibility currently rests on know-how and execution.

INFORMATION TO PREPARE

1. Which alternatives prospective customers weigh, and the criteria that decide the choice.
2. Comparative cost, performance and reliability data against the main alternatives.
3. What would prevent a larger supplier from replicating the offering, and the protection in place.

Funding history

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FINANCING POSITION

TerraSpark has completed a EUR 5m Pre-Seed round (March 2026). Reported financing totals EUR 5m Pre-Seed round (2026).

FUNDING TIMELINE

Date	Financing	Amount	Investors / providers	Key details
March 2026	Pre-Seed	EUR 5m	Daphni, Sake Bosch, better ventures, Hans(wo)men Group, Luxembourg Business Angel Network, Karaoke Club and 30 strategic business angels	Intended use: further technology development, preparation for initial pilot applications, and live tests.

Total funding EUR 5m Pre-Seed round (2026)

INVESTORS & CAPITAL PROVIDERS

Named investors include Daphni, Sake Bosch, better ventures, Hans(wo)men Group, Luxembourg Business Angel Network and Karaoke Club.

INFORMATION TO PREPARE

- 1. Current cash position, monthly burn and runway.
- 2. Any current or planned raise: amount, status, investors and intended use of the new capital.

Investor questions & information to prepare

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CONCERNS & CONFLICTS

Is any adverse event publicly reported, and where do sources disagree?

No public Concern or source Conflict is recorded in this baseline.

Part A lists the questions an investor is likely to raise after reviewing the public record; Part B lists the documents and data that evidence them. Records such as the cap table, management accounts and agreements are normally private: diligence requests them as standard, not because their absence from public sources is a concern.

A. QUESTIONS TO PREPARE FOR

No.	Information / clarification required	Why it matters
Product & technology		
1	Performance of the current product under operating conditions: reliability, performance and usage data.	Product maturity rests on company statements; operating data shows whether the product performs at customer conditions.
2	Certification or qualification required for customer operation, its current status and what remains.	Determines when the product can be sold and used.
3	What is protected: patents or applications filed, and ownership of the core technology.	No patent position is established; defensibility depends on what the company owns and can protect.
Founders, team & organisation		
4	Whether each founder works full-time for the company, given other current roles listed in public profiles (Jasper Deprez and Sanjay Vijendran), and any arrangements with those entities.	Delivery depends on a committed founding team; outside roles can also create related-party interests.
5	Current headcount by function and the key hires planned.	Shows whether the team can deliver the next milestones.
Customers & commercial traction		
6	Customer discovery to date: organisations engaged, design partners, and any letters of intent or pilots under discussion.	At pre-seed the first sign of demand is engagement from the intended buyers.
Market & go-to-market		
7	Which segment and geography the company is prioritising - among data center operators, critical infrastructure operators, defence installations and other segments - why that one first, and how many qualifying sites or customers it counts there.	Research lists several target segments; the first one sets the sales cycle, reference customers and capital need.
8	The buying process in the target segment: budget owner, decision-makers, procurement steps and time from first contact to order.	The buying cycle sets how fast revenue can follow.
Business model & economics		
9	How much revenue is recurring rather than project-based, and the margin on each.	Project revenue scales with headcount unless it becomes recurring.
Financial position		
10	Current cash, monthly burn and runway, and the capital needed to reach the next technical and commercial milestones.	For a capital-intensive business, runway against the delivery timetable sets the size and timing of the next raise.
Funding & capital structure		
11	Current fundraising: amount sought, capital already committed, use of funds, and the technical and commercial milestones it is meant to reach.	Links the investment to specific milestones.

No.	Information / clarification required	Why it matters
Competition & defensibility		
12	Alternatives actually weighed in customer discussions to date (for example continuing with terrestrial grid connection and diesel generation, or direct alternatives such as Space Solar), why the customer chose or would choose the company, what would stop a competitor replicating the offering, and any comparative cost or performance evidence.	Only buyer decisions show which alternatives compete; public research shows the field, not the customer's shortlist.

Corporate & ownership

13	Ownership of the company: founders' shareholdings, option pool, founder vesting, and rights attached to the 2026 financings.	Determines founder incentives and the rights a new investor would rank alongside.
14	The legal entity behind the company - registered name, jurisdiction and registration number - and that it is active.	Financing, contracts and intellectual property attach to a specific legal entity.

B. SUPPORTING DOCUMENTS & DATA TO PREPARE

Priority - initial review

Secondary - follow-up

Corporate, legal & ownership

- Certificate of incorporation or current registry extract and constitutional documents.
- Shareholders' agreement and any investor side letters.

- Schedule of related-party arrangements, including with entities in which founders hold roles.

Financial information

- Annual financial statements for each financial year since incorporation (2025).
- Current-year management accounts (P&L, balance sheet, cash flow), latest cash balance, monthly burn and runway.
- Budget and financial forecast to the next funding milestone, with key assumptions.
- Key metrics pack: revenue by client, backlog, and utilisation by team.

Customers & commercial

- Letters of intent, MoUs, pilot agreements or other customer commitments.
- Pipeline report (CRM extract) and order book.

- Standard commercial terms and price list.
- Agreements with named partners that define their role (Dcubed).

Product & technology

- Product and technical roadmap to the next milestones.
- Product analytics and uptime or incident reports.

- Technical architecture, with critical dependencies and third-party technology.
- Security and data-protection documentation.

Intellectual property

- Schedule of patents, applications and trademarks, or confirmation that none has been filed.
- IP assignment agreements from founders, employees and contractors.

- Schedule of material open-source and third-party software licences.

Organisation & people

- Organisation chart and headcount by function, with the hiring plan for the next 12-18 months.

- Service agreements of the founders and key staff, including arrangements covering their other current roles.
- Option plan (ESOP) and grants to date.

Funding & securities

- Fully diluted cap table.
- Signed financing agreements for each reported 2026 financing.
- Current fundraising materials (investor presentation and use-of-funds plan).

Regulatory / sector-specific

- Permits, safety certification and environmental compliance for deployments.

ABOUT THIS SCREEN

Independent public-source assessment showing how the company may initially appear to an investor, before founder clarification, supporting evidence or private diligence. It is intentionally the external, public-source view: information that public sources do not establish may still exist within the company.

- Evidence is limited to publicly observable sources at the time of research.
- This baseline is prepared from reviewed public sources before founder enrichment. Founder confirmations, corrections and supporting evidence are recorded separately and never silently overwrite it.

Research window: 16 February 2026 – 30 March 2026

SOURCES

#	SOURCE	PUBLISHED	LINK
S1	tech.eu - TerraSpark advances space-based solar power with €5M ...	30 March 2026	tech.eu/2026/03/30/terraspark-advances-space-based-solar-power-with-eur5m-funding
S2	terraspark.energy - How Does Space-Based Solar Power Work? - TerraSpark	16 February 2026	www.terraspark.energy/blog/how-does-space-based-solar-power-work
S3	terraspark.energy - Wireless Energy from Space - TerraSpark	30 March 2026	www.terraspark.energy/blog/wireless-energy-from-space
S4	terraspark.energy - TerraSpark Powering Earth from Space		www.terraspark.energy
S5	linkedin.com		linkedin.com/in/jasper-d-a780b794
S6	terraspark.energy - TerraSpark Secures €5M for Space Energy Vision	30 March 2026	www.terraspark.energy/blog/terraspark-secures-five-million-euros
S7	linkedin.com		linkedin.com/in/matthias-laug-202279226
S8	dcubed.space - Dcubed Announces Five Power Beaming Partners for ...	23 March 2026	dcubed.space/dcubed-announces-five-power-beaming-partners-for-araqys-d3-mission
S9	linkedin.com		linkedin.com/in/sanjay-vijendran-2018